

3RD HARSH-ENVIRONMENT MASS SPECTROMETRY WORKSHOP

Adaptation of a commercially available RGA for use onboard the ISS

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ABSTRACT

A residual gas analyzer (RGA) was developed for the Material Science Laboratory - Vacuum Gas System (MSL-VGS) onboard the ISS to determine krypton leakage in sealed cartridges before and during vacuum metallurgy experiments. Krypton leakage is an indication of rupture of the cartridges which may contain dangerous chemicals such as arsenic and thus the RGA is a central safety component of the metal processing facility. The RGA also allows composition analysis of off-gases and by products prior to release to the ISS vacuum lines. A commercially available RGA (PRISMA QMS 200) was selected and adapted as it offers several advantages as e.g.: a) Due to the large number of systems manufactured already (several thousands), all problems that may show up with more prototype like systems have been overcome already, b) based on a standard system, the instrument could be developed at a reasonable price, c) the instrument had flown on a previous shuttle mission and several parabolic flights. The adaptations of the standard parts for use in space included special fixation of all mechanical and bigger electronics parts to withstand the enormous forces during launch on board the space shuttle, replacement of flammable and dangerous components and special coating of all electronics boards. For mounting, transportation, dust tightness, depressurization and heat management a special housing was designed. Flight and test systems are completed and final tests have been performed so that the units can be transported to and installed in the International Space Station. The design of the instrument as well as all necessary adaptations and the results of performance tests will be presented.